SDMCNet: Spatial–Spectral Decoupled Mamba–MoE Collaborative Network for Hyperspectral Image Classification

Xue Chai, Bingbing Lei, Wei Feng, Ruiqing Zhang, and Zhipeng Liu

***Abstract*—Using the whole image as input for hyperspectral image (HSI) classification, the model needs to handle broader spatial dependencies and more diverse spectral variations. In modeling nonlocal spatial relationships, scanning based on fixed windows is prone to directional bias, leading to the loss of spatial boundary details. In addition, the high dimensionality of the data makes spectral distributions complex and spectral information redundant, thereby affecting the reliable identification and effective utilization of spectral bands. To address these challenges, we propose SDMCNet, a spatial–spectral decoupled Mamba–mixture-of-expert (MoE) collaborative network for image-level HSI classification. Taking the whole image as input, SDMCNet achieves a balance between spatial structure modeling and fine-grained spectral representation. Specifically, the spatial branch progressive spatial perception feature extraction (PSPFE) integrates local and global Mamba modules to capture short- and long-range spatial contextual dependencies. A boundary constrained enhancement module (BCEM) is further introduced to suppress oversmoothing and noisy perturbations near class boundaries, thereby preserving spatial boundary details. The spectral branch anchor routed expert spectral aggregation (ARESA) builds an anchor routed expert collaboration mechanism to adaptively select complementary spectral experts to model complex dependencies among spectral bands and obtain precise spectral band combinations. The two branches collaborate through spectral channel weighting, thereby achieving synergistic spatial–spectral modeling for image-level HSI classification. Extensive experiments on the Indian Pines (IP), Pavia University (PU), Houston 2013, and HanChuan (HC) datasets demonstrate that SDMCNet consistently outperforms representative methods, validating the effectiveness and potential value of using the whole image as input for HSI classification. The code is available at https://github.com/chai-xue/SDMCNet**

***Index Terms*—Boundary constrained enhancement, fine-grained spectral discrimination, hyperspectral image (HSI) classification, Mamba–mixture-of-expert (MoE) collaborative, spatial–spectral decoupled.**

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I. Introduction

HYPERSPECTRAL imagery provides radiance measurements with fine spectral sampling, enabling the retrieval of reflectance information for terrestrial targets and thereby substantially improving the discrimination of ground materials. Benefiting from this capability, hyperspectral image (HSI) has been widely used in agricultural monitoring [1], urban mapping [2], resource and environmental surveys [3], and mineral exploration [4]. In these applications, HSI classification typically relies on the joint exploitation of spatial context and spectral information to assign a class label to each pixel [5], thereby providing fundamental support for land-cover interpretation and downstream decision-making.

Early HSI classification primarily relied on per-pixel spectral features and conventional machine learning classifiers, such as support vector machine (SVM) [6], [7], [8] and random forest (RF) [9]. However, these methods have limited adaptability to complex nonlinear spatial–spectral relationships. With the development of convolutional neural networks (CNNs) [10], [11], [12], spatial–spectral joint feature learning has become a mainstream approach for HSI classification. Zhong et al. [13] designed the spatial–spectral residual network (SSRN) to perform joint feature learning by combining 3-D convolutions with residual connections. Roy et al. [14] proposed HybridSN, which cascades 3-D and 2-D convolutions to balance spatial–spectral coupling and spatial discrimination. Zhou et al. [15] designed attention multihop graph and multiscale convolutional fusion network (AMGCFN) to model irregular neighborhood relationships through multiscale convolution and multihop graph convolution. Although these deep models enhance the capacity for spatial–spectral feature learning, their interactions remain largely constrained by local receptive fields and limited neighborhood propagation. Consequently, these methods remain inadequate at leveraging global context and achieving robust generalization across different areas.

To enhance long-range information interaction, Transformer models based on self-attention mechanisms [16] have been introduced into HSI classification, in which tokenization enables the model to more effectively capture global dependencies. Hong et al. [17] designed SpectralFormer to model adjacent band relationships through spectral sequences. Sun et al. [18] introduced spectral–spatial feature tokenization Transformer (SSFTT), which uses spatial–spectral tokens to facilitate cross-domain information exchange and joint

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modeling. Yang et al. [19] proposed HiT, incorporating convolutional inductive biases into the Transformer architecture to balance local details and global context. Yang et al. [20] developed center-to-surrounding interactive learning (CSIL), which exploits center-to-periphery interactions and hierarchical region sampling to capture multiscale relationships. Butt et al. [21] proposed GFormer, which introduces graph structured priors into a Transformer to fuse interband relationships and spatial context. Arshad and Zhang [22] proposed a lightweight spatial–spectral Transformer that jointly models spatial and spectral dependencies via lightweight self-attention to enhance classification representations. Feng et al. [23] extended Transformer architectures to cross-domain few-shot hyperspectral analysis by transferring knowledge from source domains to target scenes under limited supervision. Jiang et al. [24] systematically reviewed cross-domain HSI classification and summarized the major challenges in cross-scene generalization. Although Transformer-based methods effectively capture global dependencies, their computational and memory costs increase rapidly with token length, limiting their application to high-resolution spatial and high-dimensional spectral modeling. Moreover, fixed token construction and attention patterns restrict their adaptability to scale variations, noise perturbations, and complex spectral distributions. Therefore, efficient and adaptive spatial–spectral modeling strategies remain necessary.

Recently, structured state-space models (SSMs) have provided an efficient route with near linear complexity for learning long-sequence dependencies. Gu et al. [25] proposed S4, which enables scalable long-range dependency modeling via structured parameterization. For nonsequential visual data, Zhu et al. [26] introduced ViM, which feeds image block sequences into bidirectional state-space layers to model global interactions. Liu et al. [27] developed VMamba with 2-D selective scanning to support sequential modeling on grid structured data. For HSI classification, He et al. [28] proposed IGroupSS-Mamba, which mitigates redundancy and improves spatial–spectral synergy via spectral band grouping and multiple directional scanning. Huang et al. [29] designed 3DSS-Mamba to strengthen spatial–spectral interactions. Lu et al. [30] proposed spatial–spectral unified Mamba (SSUM), which jointly models spectral and spatial branches with a unified SSM framework. Li et al. [31] introduced MambaHSI, which models long-range spatial–spectral relationships via whole-image interaction and adaptive fusion. However, the above methods introduce directional bias when serializing 2-D grids. This limitation weakens the discriminability of spatial boundary details, thereby reducing classification stability in complex scenes. In addition, since pixels contain multiple spectral responses, existing methods lack the ability to explicitly decouple fine-grained spectral responses.

To enhance model adaptability to complex spatial–spectral relationships, mixture of experts (MoE) has been introduced into HSI classification. He et al. [32] proposed HSI-MFormer, which employs Transformer experts for feature interaction within each level and Mamba experts to exchange information across feature levels, combining short-range self-attention with long-range state-space modeling. Dai et al. [33] proposed HyperTransXNet, which uses sparse routing to enhance spatial and spectral representations. These studies demonstrate the value of expert collaboration for adaptive feature modeling in HSI classification. However, existing MoE-based methods mainly distinguish experts by network architecture or scale path rather than organizing them according to different spectral band modeling requirements. As a result, they cannot adapt to the continuous correlations between adjacent bands, correlations between nonadjacent bands, local absorption variations, and high-dimensional redundancy exhibited by hyperspectral signatures. Consequently, there is still no explicit expert collaboration mechanism for complementary modeling of different types of spectral relationships across all spectral bands.

To address this gap, studies that use MoE in other remote sensing tasks to improve model performance provide valuable guidance for our work [34], [35], [36]. Wang et al. [34] proposed ME-SSM for optical and synthetic aperture radar (SAR) image registration, in which different experts process views generated by different transformations, and a soft router dynamically aggregates their responses, demonstrating that coupling representations from multiple views can improve feature discrimination. Kong et al. [35] proposed CPLP-DMoE for joint HSI and LiDAR classification, using band mixture and multimodal fusion experts to address adjacent band similarity and the limitations of a single fusion strategy, demonstrating that different representation demands can be handled by specialized experts. Lin et al. [36] proposed RS-MoE for remote sensing image captioning and visual question answering and enabled task-related expert selection through an instruction router rather than relying on a fixed expert combination. Inspired by these studies, we construct a complementary spectral modeling MoE mechanism and coordinate multiple experts through sparse routing for global spectral feature enhancement.

Against this background, we propose SDMCNet, a spatial–spectral decoupled Mamba–MoE collaborative network for HSI classification with whole-image input. Rather than coupling spatial and spectral cues within a single feature extractor, SDMCNet introduces Mamba into the spatial branch to model long-range spatial contextual dependencies and MoE into the spectral branch to route different spectral responses to specialized experts, thereby supporting fine-grained spectral representations under the influence of complex spectral distributions and spectral redundancy. This design further enables efficient spatial–spectral collaborative modeling within a decoupled framework. Specifically, the spatial branch, progressive spatial perception feature extraction (PSPFE), combines the dual-stream local refinement module (DLRM) and MultiPath scan Mamba module (MPSM) to build local-to-global spatial contextual representations, with multipath Mamba modeling mitigating directional bias. To address boundary continuity issues, a boundary-constrained enhancement module (BCEM) is designed to regulate feature updates around class boundaries, thereby preserving boundary details. The spectral branch, anchor routed expert spectral aggregation (ARESA), employs selective spectral anchor grouping (SSAG) to construct spectral groups through anchor-guided deformable sampling, thereby emphasizing discriminative spectral responses. Given the spectral distribution

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characteristics of each group, the long-range dynamic channel cluster (LRDCC) employs MoE-based sparse routing to select an appropriate combination of spectral experts, thereby obtaining refined spectral representations. The resulting spectral information recalibrates spatial structural features through spectral channel weighting, thereby establishing a complementary spatial–spectral modeling mechanism for accurate classification.

To summarize, the main contributions of this article are as follows.

1) We propose SDMCNet, a spatial–spectral decoupled Mamba–MoE collaborative network. Given a whole image as input, it jointly models spatial structural information and fine-grained spectral representations, thereby achieving efficient modeling of spatial–spectral complementarity for image-level classification in complex scenes.
2) The spatial branch PSPFE is designed with Mamba-based local feature enhancement and multipath scanning interactions to model local and global spatial contextual dependencies while mitigating directional bias. A boundary constrained multiscale correction strategy is proposed to enhance the model's discriminative capability at class boundaries.
3) The spectral branch ARESA employs anchor-guided deformable sampling to capture more discriminative spectral group responses. The MoE-based sparse routing strategy is introduced to coordinate complementary experts in modeling complex intrinsic spectral dependencies, thereby enhancing the network's refined representation of spectral bands.
4) Extensive experiments are conducted on four hyperspectral benchmark datasets. The results demonstrate the effectiveness and robustness of the proposed method in complex settings.

The remainder of this article is organized as follows. Section II describes the proposed SDMCNet framework. Section III presents the experimental setup and analyzes the results. Section IV examines key design choices and model behavior. Finally, Section V concludes this article and outlines future work.

II. Methodology

A. Preliminaries

1) SSMs and Mamba: SSMs represent sequential data through latent state evolution. Given an input sequence $x(t) \in \mathbb{R}$, an N-dimensional latent state $h(t) \in \mathbb{R}^N$, and an output $y(t) \in \mathbb{R}$, a continuous SSM is formulated as

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N\times N}$ is the state transition matrix and $\mathbf{B} \in \mathbb{R}^{N\times 1}$ and $\mathbf{C} \in \mathbb{R}^{1\times N}$ are the input and output projection matrices, respectively.

To support discrete sequence modeling, (1) is discretized using a zero-order hold with a discretization step Δ, yielding

$$\begin{aligned} h_t &= \overline{\mathbf{A}}\, h_{t-1} + \overline{\mathbf{B}}\, x_t \\ y_t &= \mathbf{C}\, h_t \end{aligned} \tag{2}$$

where $\overline{\mathbf{A}} = \exp(\Delta\mathbf{A})$ and $\overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}\big(\exp(\Delta\mathbf{A}) - \mathbf{I}\big) \cdot \Delta\mathbf{B} \approx \Delta\mathbf{B}$. The recurrence in (2) can be equivalently expressed as a structured convolution

$$\begin{aligned} \overline{\mathbf{K}} &= \left(\mathbf{C}\overline{\mathbf{B}}, \mathbf{C}\overline{\mathbf{A}\mathbf{B}}, \ldots, \mathbf{C}\overline{\mathbf{A}}^{L-1}\overline{\mathbf{B}}\right) \\ \mathbf{y} &= \mathbf{x} * \overline{\mathbf{K}} \end{aligned} \tag{3}$$

where L is the sequence length, $\overline{\mathbf{K}} \in \mathbb{R}^L$ is the convolution kernel, and $*$ denotes discrete convolution. This equivalence enables S4 [25] to efficiently model long-range dependencies with $O(L)$ complexity.

Traditional SSMs achieve linear-time computation but are limited in capturing content variations across sequence positions. To overcome this limitation, Mamba [37] introduces the selective SSM (S6) to enable content-aware sequence modeling. Specifically, given an input $x \in \mathbb{R}^{B\times L\times D}$, where B, L, and D denote the batch size, sequence length, and feature dimension, respectively, the corresponding formulation is given by

$$\begin{aligned} \mathbf{B} &= \text{Projection}_B(x) \\ \mathbf{C} &= \text{Projection}_C(x) \\ \Delta &= \tau_\Delta\left(\text{Parameter} + \text{Projection}_\Delta(x)\right) \end{aligned} \tag{4}$$

where $\mathbf{B} \in \mathbb{R}^{B\times L\times N}$, $\mathbf{C} \in \mathbb{R}^{B\times L\times N}$, and $\Delta \in \mathbb{R}^{B\times L\times D}$ are generated from x through linear projections and nonlinear transformations, enabling position-wise modulation of state updates and output readouts while preserving the linear recurrence in (2). With a hardware-aware selective scan, Mamba maintains linear time and space complexity while enhancing contextual dependency modeling.

2) Mixture of Expert: MoE distributes model capacity across multiple expert subnetworks and uses a gating network to perform expert routing. Given input features $\mathbf{v}$, the gating function $z(\mathbf{v})$ generates expert routing weights $\mathbf{q}(\mathbf{v})$, and the MoE output is formulated as

$$\mathbf{h}(\mathbf{v}) = \sum_{e=1}^{E} q_e(\mathbf{v})\, \mathcal{E}_e(\mathbf{v}), \quad \mathbf{q}(\mathbf{v}) = \text{Softmax}(z(\mathbf{v})) \tag{5}$$

where $\{\mathcal{E}_e\}_{e=1}^{E}$ denotes the set of expert subnetworks with distinct inductive biases and $q_e(\mathbf{v})$ denotes the routing weight assigned to expert e.

To limit the computational cost per sample, sparse MoE [38] adopts Top-k routing, which selects only k experts for each input

$$\mathcal{S}(\mathbf{v}) = \text{TopK}(\mathbf{q}(\mathbf{v}), k), \quad \mathbf{h}(\mathbf{v}) = \sum_{e\in\mathcal{S}(\mathbf{v})} q_e(\mathbf{v})\, \mathcal{E}_e(\mathbf{v}) \tag{6}$$

where $\mathcal{S}(\mathbf{v})$ denotes the selected expert set. This design allows the model capacity to increase with the number of experts while limiting the computation for each sample to k experts.

B. Overall Architecture

To enable joint modeling of spatial nonlocal correlations and fine-grained spectral discrimination for whole image inputs, we propose SDMCNet, whose overall architecture is shown in Fig. 1. It comprises the spatial branch PSPFE and the

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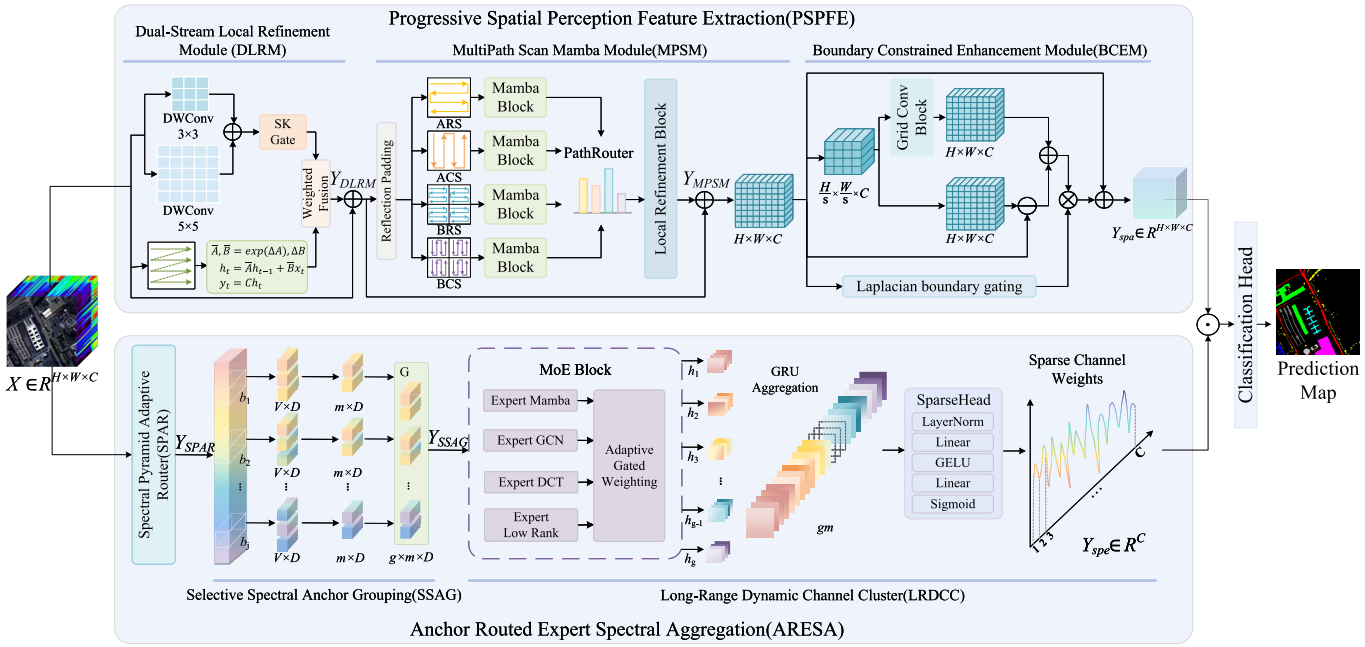


Fig. 1. Overview of the proposed SDMCNet framework.

spectral branch ARESA, which together form a decoupled classification framework. The spatial branch is designed as a cascaded structure based on Mamba, in which the DLRM introduces local priors to enhance spatial representations, whereas the MPSM captures long-range contextual dependencies through a multipath scanning interaction mechanism. In addition, the BCEM leverages multiscale contextual information to strengthen class boundary perception and preserve structural details. The spectral branch employs the spectral pyramid adaptive router (SPAR) to capture short- and medium-range spectral correlations and the SSAG to reconstruct the long spectral sequence into spectral groups that focus on key responses. The LRDCC performs sparse routing based on MoE to match each group with complementary experts according to its characteristics and uses a gated recurrent unit (GRU) for long-range recursive aggregation of channels across groups. The spectral branch provides complementary discriminative spectral representations for the spatial branch, and the two branches work collaboratively to achieve spatial–spectral joint modeling, thereby improving whole-image classification performance on HSI.

The input feature map of SDMCNet is $X \in \mathbb{R}^{H\times W\times C}$, where H and W denote the spatial dimensions and C denotes the number of spectral channels. The input is fed in parallel into the spatial branch PSPFE and the spectral branch ARESA, and the resulting features are fused for subsequent classification.

In PSPFE, DLRM is first employed to perform feature embedding and local short-range modeling on the input features. Then, MPSM is employed to model long-range spatial contextual dependency relationships over the entire feature map while alleviating the bias induced by fixed scanning directions, which can be formulated as

$$Y_{\mathrm{DLRM}} = \mathrm{DLRM}(X) \tag{7}$$

$$Y_{\mathrm{MPSM}} = \mathrm{MPSM}(Y_{\mathrm{DLRM}}). \tag{8}$$

Building on this, BCEM suppresses unstable updates near class boundaries to preserve boundary details, yielding the final spatial representation

$$Y_{\mathrm{spa}} = \mathrm{BCEM}(Y_{\mathrm{MPSM}}). \tag{9}$$

$Y_{\mathrm{spa}} \in \mathbb{R}^{H\times W\times C}$ preserves the original spatial resolution while encoding multiscale spatial features that combine global structural consistency with local boundary sensitivity.

In ARESA, SPAR first normalizes and compresses the spatial responses of each channel to generate spectral token representations

$$Y_{\mathrm{SPAR}} = \mathrm{SPAR}(X) \tag{10}$$

where $Y_{\mathrm{SPAR}} \in \mathbb{R}^{C\times D}$ and D denotes the feature dimension of each spectral token.

Based on Y_{SPAR}, SSAG selects anchors along the channel dimension and predicts deformable offsets within local neighborhoods, thereby sparsifying dense spectral sequences into group-level token representations. Subsequently, LRDCC uses multiple lightweight experts to dynamically aggregate the spectral tokens within each group and employs a recursive unit to model long-range dependencies between groups, ultimately generating sparse channel weights

$$Y_{\mathrm{SSAG}} = \mathrm{SSAG}(Y_{\mathrm{SPAR}}) \tag{11}$$

$$Y_{\mathrm{spe}} = \mathrm{LRDCC}(Y_{\mathrm{SSAG}}) \tag{12}$$

where $Y_{\mathrm{spe}} \in \mathbb{R}^{C}$, $Y_{\mathrm{SSAG}} \in \mathbb{R}^{g\times m\times D}$, g is the number of groups, and m is the number of tokens within each group.

Y_{spe} characterizes the relative importance of different spectral bands for the current sample along the channel dimension and provides a basis for modeling spectral uncertainty in subsequent adaptive head fusion.

Finally, the spatial and spectral branches are coupled via channel-wise modulation. After broadcasting Y_{spe} over the

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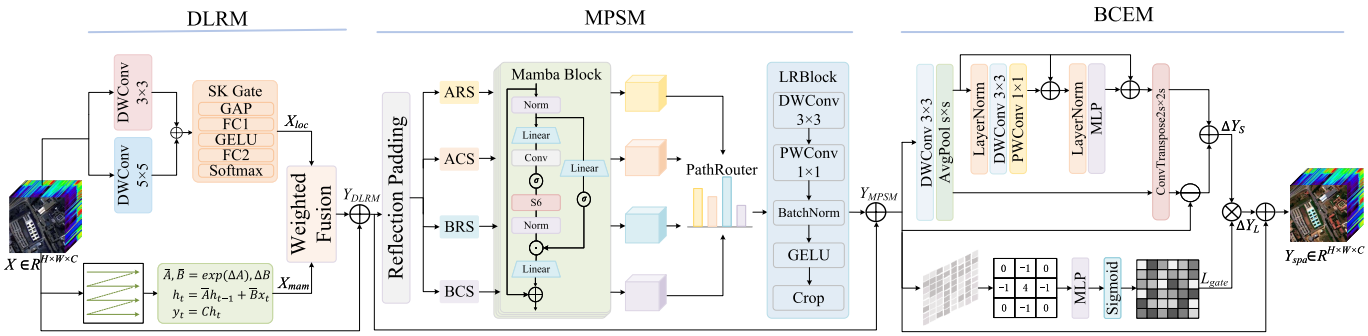


Fig. 2. Graphical illustration of the proposed PSPFE spatial branch.

spatial dimensions, we perform element-wise multiplication with the spatial feature Y_{spa} to obtain the fused representation

$$Y_{\text{out}} = Y_{\text{spa}} \odot Y_{\text{spe}} \tag{13}$$

where $Y_{\text{out}} \in \mathbb{R}^{H\times W\times C}$ and $\odot$ denotes the element-wise multiplication.

Based on the fused feature Y_{out}, we employ two classification heads, namely a main head and a prototype head, to generate pixel-wise logits

$$\begin{aligned} F_{\text{main}} &= \text{Head}_{\text{main}}\left(Y_{\text{out}}\right) \\ F_{\text{proto}} &= \text{Head}_{\text{proto}}\left(Y_{\text{out}}\right) \end{aligned} \tag{14}$$

where $F_{\text{main}}, F_{\text{proto}} \in \mathbb{R}^{H\times W\times K}$, in which K denotes the number of classes. The main classification head directly leverages the discriminative capacity of convolutional features, whereas the prototype head promotes intraclass clustering through class prototype constraints. To enable sample adaptive fusion, a coefficient $\xi \in [0,1]$ is derived from the entropy of the normalized channel weights and used to combine the outputs of the two heads. Finally, the predicted class of each pixel is obtained by applying argmax

$$F_{\text{out}} = (1-\xi)\, F_{\text{main}} + \xi F_{\text{proto}} \tag{15}$$

$$\hat{Y} = \arg\max\left(F_{\text{out}}\right) \tag{16}$$

where $F_{\text{out}} = [F_1, F_2, \ldots, F_K] \in \mathbb{R}^{H\times W\times K}$ denotes the fused logits and $\hat{Y} \in \mathbb{R}^{H\times W}$ denotes the predicted label map.

C. Progressive Spatial Perception Feature Extraction

PSPFE adopts a progressive spatial modeling strategy to refine spatial representations while preserving structural details, as illustrated in Fig. 2.

1) Dual-Stream Local Refinement Module: DLRM combines multiscale depth-wise separable convolutions with Mamba blocks to preserve 2-D spatial structure while simultaneously enhancing local detail features and modeling the corresponding spatial context. This design produces spatial features that fuse local structure with neighborhood dependencies.

The local branch uses dual-scale depth-wise separable convolutions to extract neighborhood responses and adaptively fuses the two branches via an SK Gate to form local spatial features

$$X_{\text{loc}} = \text{SK}\left(\text{DWConv}_{3\times 3}\left(X\right), \text{DWConv}_{5\times 5}\left(X\right)\right). \tag{17}$$

The Mamba branch flattens X into a sequence $T = \text{flatten}(X) \in \mathbb{R}^{L\times C}$, where $L = HW$. After normalization and linear mapping, it is split into two paths: convolution update and gating. This yields short-range dependency responses, which are then rearranged back into the 2-D spatial structure to produce the Mamba branch output

$$\begin{aligned} T_x,\, T_z &= \text{chunk}\big(\text{Linear}\left(\text{LN}\left(T\right)\right)\big) \\ T_m &= \text{S6}\big(\sigma\left(\text{Conv}\left(T_x\right)\right)\big) \\ X_{\text{mam}} &= \text{Reshape}\left(T + \text{Linear}\left(\text{LN}\left(T_m\right)\right) \odot \sigma\left(T_z\right)\right). \end{aligned} \tag{18}$$

The outputs from the convolution branch and Mamba branch are linearly fused in the channel dimension via scalar weights w_{loc} and w_{mam}. The final DLRM output is obtained through a channel feedforward network with residual connections

$$Y_{\text{DLRM}} = X + \eta\left(w_{\text{loc}} X_{\text{loc}} + w_{\text{mam}} X_{\text{mam}}\right) \tag{19}$$

where $Y_{\text{DLRM}} \in \mathbb{R}^{H\times W\times C}$ and η is the residual scaling coefficient.

2) MultiPath Scan Mamba Module: To enhance nonlocal modeling over the entire image and mitigate directional bias induced by single-path scanning, the spatial branch further incorporates the MPSM module. Multiple serpentine scanning paths are constructed on the same input feature map and processed by the S6 module to learn richer nonlocal spatial dependencies. The feature map of Y_{DLRM} after reflection padding is denoted as $Y_z \in \mathbb{R}^{H_z\times W_z\times C}$. Continuous serpentine paths are constructed by alternating traversal along the row and column dimensions. In addition, the spatial region is partitioned into blocks of size $z \times z$ to form two block-level path types, namely block row serpentines and block column serpentines. For any path $a \in \{\text{ARS}, \text{ACS}, \text{BRS}, \text{BCS}\}$, the corresponding sequence can be written as

$$T_a = S_a\left(Y_z\right) \tag{20}$$

where $T_a \in \mathbb{R}^{L\times C}$ with $L = H_z \times W_z$ and $S_a(\cdot)$ denotes the serpentine scan transformation corresponding to path a. Each sequence undergoes Mamba encoding with identical architecture but independent parameters, yielding four spatial responses

$$M_a = \text{Mamba}_a\left(T_a\right) \tag{21}$$

where $M_a \in \mathbb{R}^{H_z\times W_z\times C}$ and Mamba$(\cdot)$ is defined in (18). To enable adaptive path selection, a lightweight path router

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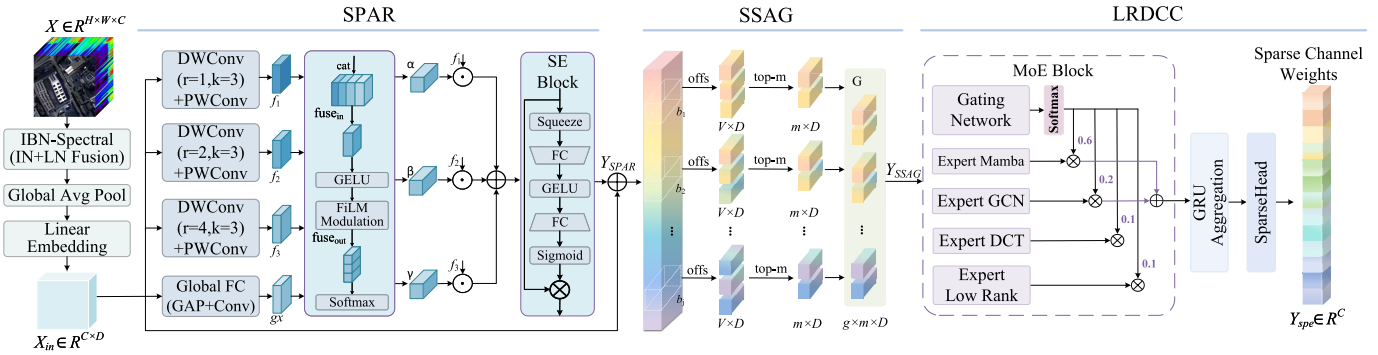


Fig. 3. Graphical illustration of the proposed ARESA spectral branch.

is employed. For any path M_a, global channel statistics are extracted and processed through a feedforward network with softmax to obtain normalized weights w_a. During training, one path is randomly masked to prevent the model from over-relying on a single path. Finally, the four responses are combined by weighted summation to yield the fused long-range spatial features

$$Y_{\text{fuse}} = \sum_a w_a \,\text{PathDrop}\,(M_a)\,. \tag{22}$$

The fused features are refined by the repair unit LRB$(\cdot)$ and then added to the residual input to produce the output of the MPSM module

$$Y_{\text{MPSM}} = Y_{\text{DLRM}} + \text{LRB}\,(Y_{\text{fuse}}) \tag{23}$$

where $Y_{\text{MPSM}} \in \mathbb{R}^{H\times W\times C}$.

3) Boundary Constrained Enhancement Module: Multiscale feature reconstruction and boundary-aware gating are jointly employed to selectively suppress unstable updates near class boundaries.

The input features are smoothed within local neighborhoods and downsampled at scales $s \in \{2, 4\}$ to produce low-resolution feature maps

$$Y^s_{\text{grid}} = \text{AvgPool}_{s\times s}\big(\text{DWConv}_{3\times 3}\,(Y_{\text{MPSM}})\big) \tag{24}$$

where $Y_{\text{grids}} \in \mathbb{R}^{H_s\times W_s\times C}$, $H_s = H/s$, and $W_s = W/s$. At scale s, we reconstruct features from Y^s_{grid} using the Grid Conv Block $G_s(\cdot)$ and use ConvTranspose 2 $\times$ 2 s to implement the upsampling operator $U_s(\cdot)$, thereby restoring the original resolution. To suppress the detail degradation caused by oversmoothing, we define the multiscale correction term as

$$\Delta Y_S = \sum_s \left[U_s\left(G_s\left(Y^s_{\text{grid}}\right)\right) + \phi\left(Y_{\text{MPSM}} - U_s\left(Y^s_{\text{grid}}\right)\right)\right] \tag{25}$$

where $\Delta Y_S \in \mathbb{R}^{H\times W\times C}$ and ϕ denotes the gain coefficient of the detail residual term, which controls the strength of high-frequency compensation.

To prevent multiscale corrections from introducing unstable perturbations near class boundaries, BCEM constructs a boundary-aware gating function. The response obtained by channel averaging is filtered by a fixed Laplacian operator and mapped through a 1 $\times$ 1 convolution with sigmoid activation to obtain $L_{\text{gate}} \in [0, 1]^{H\times W}$. With θ controlling the boundary suppression strength, the constrained multiscale correction is formulated as

$$\Delta Y_L = \left(1 - \theta L_{\text{gate}}\right) \odot \Delta Y_S \tag{26}$$

where $\odot$ denotes element-wise multiplication broadcast along the channel dimension. Finally, the boundary constrained correction is scaled by the learnable residual coefficient μ and added to the input feature

$$Y_{\text{spa}} = Y_{\text{MPSM}} + \mu\Delta Y_L. \tag{27}$$

D. Anchor Routed Expert Spectral Aggregation

ARESA combines anchor-based spectral grouping with expert collaboration to learn fine-grained discriminative spectral representations under heterogeneous spectral variations, as illustrated in Fig. 3.

1) Spectral Pyramid Adaptive Router: Before anchor grouping, SPAR combines multiscale spectral convolutions with global conditional gating to strengthen adjacent band interactions and suppress redundant responses, yielding spectral representations enriched with multiscale context for subsequent anchor routing.

We first compress spectral features into spectral tokens using IBN-Spectral [39] and spatial global average pooling (GAP), and then project them into a D-dimensional subspace

$$X_{\text{in}} = \text{Linear}\big(\text{GAP}_{H\times W}(\text{IBN}\,(X))\big) \tag{28}$$

where $X_{\text{in}} \in \mathbb{R}^{C\times D}$. Subsequently, X_{in} is transposed to $\widetilde{X}_{\text{in}} \in \mathbb{R}^{D\times C}$. A multiscale depth-wise separable convolutional branch is then built along the spectral dimension with dilation rates $r \in \{1, 2, 4\}$, producing three responses $\{f_1, f_2, f_3\} \in \mathbb{R}^{D\times C}$. Meanwhile, a conditional vector $gx \in \mathbb{R}^{D\times C}$ is derived from the global spectral statistics of $\widetilde{X}_{\text{in}}$, and they are passed through a lightweight gating network to output the multiscale weights

$$[\alpha, \beta, \gamma] = \text{softmax}\big(\Psi\,(f_1, f_2, f_3, gx)\big) \tag{29}$$

where $\Psi(\cdot)$ denotes a composite operator comprising concatenation, 1 $\times$ 1 convolution, Gaussian error linear unit (GELU), and FiLM modulation. The scale weights are used to perform a weighted fusion of multiscale responses. The fused response is then transposed back to $\mathbb{R}^{C\times D}$ to ensure dimensional consistency with the input spectral tokens. Finally, we inject the SE reweighted gated fusion features into the input

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spectral tokens using the residual scaling coefficient ρ, which controls the contribution of the residual branch

$$Y_{\mathrm{SPAR}} = X_{\mathrm{in}} + \rho\,\mathrm{SE}\left(\alpha \odot f_1 + \beta \odot f_2 + \gamma \odot f_3\right). \tag{30}$$

2) Selective Spectral Anchor Grouping: Local spectral variations may cause the positions of key responses to shift across neighboring bands, so grouping based on fixed windows may retain redundant bands or miss informative responses. Therefore, SSAG learns an offset for each anchor and adaptively samples and aggregates neighboring spectral responses to construct spectral groups focused on key responses.

We uniformly sample anchors along the spectral dimension with stride d

$$b_j = jd, \quad j = 0, \ldots, g-1, \quad g = \left\lceil \frac{C}{d} \right\rceil. \tag{31}$$

Using the spectral embedding $Z_{b_j} = Y_{\mathrm{SPAR}}[b_j] \in \mathbb{R}^D$ at each anchor point as the local sampling basis, two lightweight prediction heads generate V deformable sampling coordinates p_j and their corresponding importance scores π_j within a radius R

$$p_j = b_j + R\,\tanh\left(W_\Delta Z_{b_j}\right), \quad \pi_j = W_\pi Z_{b_j} \tag{32}$$

where $p_j, \pi_j \in \mathbb{R}^V$. We perform differentiable 1-D interpolation sampling at coordinates p_j along the spectral dimension to obtain the candidate token set

$$P_j = Q\left(Y_{\mathrm{SPAR}},\, p_j\right) \tag{33}$$

where $P_j \in \mathbb{R}^{V\times D}$ and $Q(\cdot)$ performs differentiable sampling through linear interpolation between $\lfloor p_{j,v} \rfloor$ and $\lceil p_{j,v} \rceil$. For each spectral anchor, offset coordinates are used to sample band tokens and construct an anchor-guided spectral group, rather than prediction slots as in detection Transformer (DETR). This formulation allows the sampling positions to adapt to shifts in informative spectral responses within local band neighborhoods.

Subsequently, we select the top-m candidates from P_j according to π_j to form spectral tokens

$$G_j = \mathrm{top}m\left(P_j, \pi_j\right) \tag{34}$$

where $G_j \in \mathbb{R}^{m\times D}$. Stacking the anchor-wise outputs in band indexed order yields Y_{SSAG}. Through anchor-guided deformable sampling and top-m selection, the resulting representation retains locally discriminative spectral cues while reducing the influence of redundant neighboring responses, thereby providing more reliable inputs for the subsequent expert routing in LRDCC.

3) Long-Range Dynamic Channel Cluster: Spectral curves generally exhibit marked heterogeneity across wavelength intervals, making a single representation insufficient to capture the diverse spectral characteristics in HSI data. To this end, LRDCC constructs four heterogeneous spectral experts based on inherent spectral properties and achieves dynamic aggregation of spectral channels through recurrent intergroup sequence reconstruction. Specifically, Mamba models ordered dependencies across the wavelength sequence through SSM-based sequential modeling. The graph uses dynamically learned connections to aggregate correlated responses across nonadjacent bands. Discrete cosine transform (DCT) maps spectral responses into the frequency domain to characterize variations at different frequency scales. LowRank reduces redundancy in spectral responses while preserving the dominant global dependencies across bands. Based on the complementary modeling capabilities of the four experts for different spectral structures, the gating mechanism selects a suitable expert combination according to the characteristics of each spectral group, thereby forming the expert set oriented toward multiple spectral relationships.

For the jth spectral group $G_j \in \mathbb{R}^{m\times D}$, the four experts designed for complementary spectral modeling requirements constitute the candidate expert set. Following the sparse routing mechanism in (5) and (6), the gating network selects the Top-k experts for each spectral group and aggregates their outputs to obtain the intragroup response:

$$u_j = \sum_{e\in\mathcal{S}(G_j)} q_e\left(G_j\right)\,\mathcal{E}_e\left(G_j\right). \tag{35}$$

This design allows each spectral group to select complementary expert representations according to its local spectral characteristics.

We stack all intragroup responses in anchor index order to form $U \in \mathbb{R}^{g\times m}$. Then, a GRU is applied to preserve sequential consistency across spectral groups, and a SparseHead is used to generate the final sparse channel weights

$$Y_{\mathrm{spe}} = \mathrm{SparseHead}\big(\mathrm{GRU}\left(U\right)\big). \tag{36}$$

E. Loss Function

To jointly ensure effective supervision for discriminative learning and controllable routing behavior, we adopt a composite objective that combines a masked cross-entropy loss with several regularization terms.

Masked cross entropy provides per-pixel classification supervision for the fused logits F. Let N denote the total number of spatial locations and K the number of classes. For the nth location, the corresponding logits are denoted as $F_n = [F_{n,1}, F_{n,2}, \ldots, F_{n,K}]$, and the prediction probabilities are given by

$$p_{n,k}^{\mathrm{cls}} = \frac{\exp\left(F_{n,k}\right)}{\sum_{k'=1}^{K}\exp\left(F_{n,k'}\right)}. \tag{37}$$

Let $w_n \in \{0,1\}$ indicate whether the nth location is included in the training mask, and let $y_n \in \{1,2,\ldots,K\}$ denote its ground truth label when $w_n = 1$. The masked cross-entropy loss is formulated as

$$\mathcal{L}_{\mathrm{mCE}} = -\frac{\sum_{n=1}^{N} w_n \log p_{n,y_n}^{\mathrm{cls}}}{\sum_{n=1}^{N} w_n + \varepsilon} \tag{38}$$

where ε is a small constant for numerical stability.

To avoid sampling position collapse and unstable jumps across anchors, SSAG introduces coverage and smoothness constraints to regularize the deformable sampling coordinates during training. For anchor j, the candidate sampling coordinate indexed by v is denoted as $p_{j,v}$. The coverage term is constructed using a triangular kernel

$$\mathrm{tri}\left(c, p_{j,v}\right) = \max\left(0,\, 1 - \left|c - p_{j,v}\right|\right). \tag{39}$$

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The triangular kernel measures the local contribution of each sampled spectral coordinate to nearby band positions, and the occupancy at spectral position c is defined by summing the local contributions over all anchors and candidate sampling coordinates

$$\mathrm{occ}_n(c) = \sum_{j=1}^{g}\sum_{v=1}^{V} \mathrm{tri}\left(c, p_{j,v}\right) \tag{40}$$

$$q_n(c) = 1 - \exp\left(-\zeta\, \mathrm{occ}_n(c)\right) \tag{41}$$

where $\zeta > 0$ controls the saturation rate of the occupancy response.

The coverage loss is

$$\mathcal{L}_{\mathrm{cover}} = 1 - \frac{1}{C}\sum_{c=1}^{C} q_n(c). \tag{42}$$

The smoothness constraint is defined as the discrepancy between candidate coordinates of adjacent anchors

$$\mathcal{L}_{\mathrm{smooth}} = \frac{1}{(g-1)\,V}\sum_{j=1}^{g-1}\sum_{v=1}^{V}\left|p_{j+1,v} - p_{j,v}\right|. \tag{43}$$

The overall SSAG regularization is defined as

$$\mathcal{L}_{\mathrm{reg}} = 10^{-3}\mathcal{L}_{\mathrm{smooth}} + 5\times 10^{-4}\mathcal{L}_{\mathrm{cover}}. \tag{44}$$

Together with deformable anchor sampling, the coverage and smoothness constraints regularize the sampling coordinate distribution, thereby alleviating sampling collapse and abrupt transitions across adjacent anchor groups while improving the continuity of reconstructed spectral groups and the compatibility of subsequent expert responses.

To control routing uncertainty and discourage excessively stochastic routing, we introduce an entropy regularizer on the routing probability vector $p_n^{\mathrm{gate}} = (p_{n,1}^{\mathrm{gate}}, p_{n,2}^{\mathrm{gate}}, \ldots, p_{n,E}^{\mathrm{gate}})^{\top}$. Minimizing this entropy reduces routing uncertainty

$$\mathcal{L}_{\mathrm{ent}} = -\frac{1}{N}\sum_{n=1}^{N}\sum_{e=1}^{E} p_{n,e}^{\mathrm{gate}} \log\left(p_{n,e}^{\mathrm{gate}} + \varepsilon\right). \tag{45}$$

To further improve training stability and expert coordination in terms of division of labor, we introduce an MoE auxiliary loss consisting of load balancing and expert diversity regularizers. The load balancing regularization is defined as

$$\mathcal{L}_{\mathrm{lb}} = \sum_{e=1}^{E}\overline{p}_e \log\left(\overline{p}_e + \varepsilon\right), \quad \overline{p}_e = \frac{1}{N}\sum_{n=1}^{N} p_{n,e}^{\mathrm{gate}}. \tag{46}$$

This term promotes a more uniform distribution of $\{\overline{p}_e\}_{e=1}^{E}$, thereby mitigating imbalance in expert usage. Furthermore, to reduce the similarity across experts, we introduce a diversity regularizer. Let the output vector of expert e for group j be $o_e^j \in \mathbb{R}^m$, which is normalized as $\tilde{o}_e^{(j)} = o_e^{(j)}/(\|o_e^{(j)}\|_2 + \varepsilon)$. The diversity loss is defined as the average pair-wise similarity across all expert pairs

$$\mathcal{L}_{\mathrm{div}} = \frac{2}{E(E-1)}\sum_{1\le a<b\le E}\frac{1}{g}\sum_{j=1}^{g}\frac{1}{m}\sum_{t=1}^{m}\tilde{o}_{a,t}^{(j)}\tilde{o}_{b,t}^{(j)}. \tag{47}$$

The resulting MoE auxiliary loss is

$$\mathcal{L}_{\mathrm{moe}} = 10^{-3}\mathcal{L}_{\mathrm{lb}} + 10^{-3}\mathcal{L}_{\mathrm{div}}. \tag{48}$$

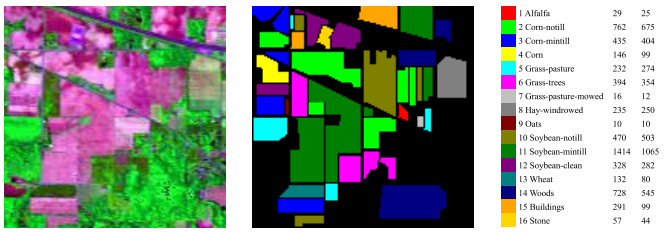


Fig. 4. IP dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.

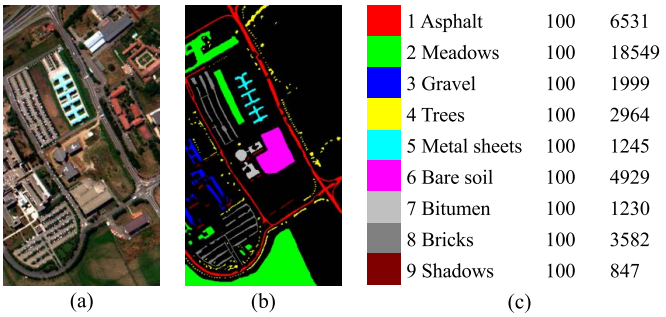


Fig. 5. PU dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.

The overall training objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{mCE}} + \lambda_{\mathrm{ent}}\mathcal{L}_{\mathrm{ent}} + \lambda_{\mathrm{moe}}\mathcal{L}_{\mathrm{moe}} + \lambda_{\mathrm{reg}}\mathcal{L}_{\mathrm{reg}} \tag{49}$$

where λ_{ent}, λ_{moe}, and λ_{reg} are scalar weighting coefficients.

III. Experiments

In this section, we first present the experimental setup. We then conduct a comprehensive evaluation through quantitative and qualitative comparisons with representative methods. Finally, ablation studies are conducted to verify the effectiveness of the proposed modules.

A. Experimental Datasets

1) Indian Pines: The Indian Pines (IP) dataset, acquired by airborne visible/infrared imaging spectrometer (AVIRIS) over northwestern Indiana, USA, in 1992, comprises 145 × 145 pixels with a spatial resolution of 20 m and 200 usable bands spanning 400–2500 nm. It contains 10 400 labeled pixels from 16 classes, with the false color image, ground truth map, and sample split shown in Fig. 4.

2) Pavia University: The Pavia University (PU) dataset was acquired by the airborne ROSIS-03 sensor over Pavia, Italy, in 2003. It comprises 610 × 340 pixels with a spatial resolution of 1.3 m and 103 spectral bands spanning 430–860 nm. The dataset contains 42 776 labeled pixels from nine classes, as shown in Fig. 5.

3) Houston 2013: The Houston 2013 (UH) dataset was acquired by the ITRES CASI-1500 sensor over the University of Houston campus and surrounding areas in Texas, USA. It comprises 349 × 1905 pixels with a spatial resolution of 2.5 m and 144 spectral bands spanning 380–1050 nm. The

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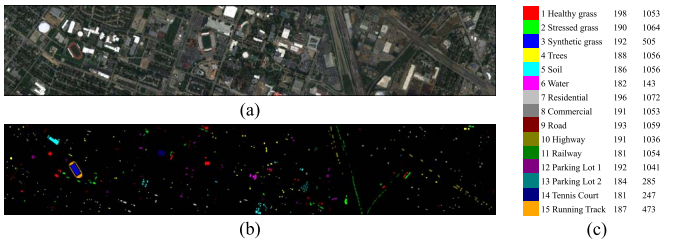


Fig. 6. Houston 2013 dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.

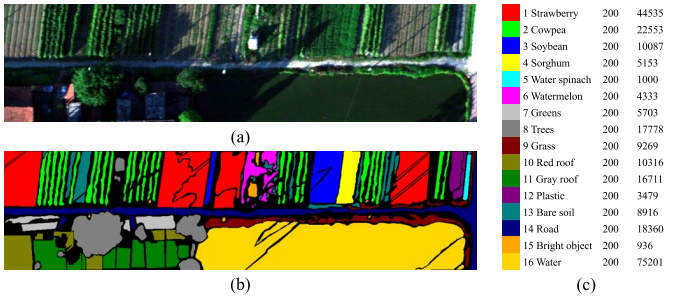


Fig. 7. HC dataset. (a) False color image. (b) Ground truth. (c) Category and sample settings.

dataset contains 15 029 labeled pixels from 15 classes, as shown in Fig. 6.

4) HanChuan: The HanChuan (HC) dataset was acquired by a Headwall Nano-Hyperspec sensor mounted on a Leica Aibot X6 UAV V1 over HC City, Hubei Province, China. It comprises 1217 × 303 pixels with an approximate spatial resolution of 0.109 m and 274 spectral bands spanning 400–1000 nm. The dataset contains approximately 257 530 labeled pixels from 16 classes, as shown in Fig. 7.

B. Experimental Setup

1) Evaluation Metrics: To effectively evaluate the quantitative performance of HSI classification models, we adopt four widely used metrics for assessing classification accuracy, including overall accuracy (OA), average accuracy (AA), the kappa coefficient (κ), and category accuracy (CA). Together, these four metrics comprehensively measure the practical performance of the model from different perspectives, ensuring that the evaluation results are objective and reliable.

2) Implementation Details: All experiments were implemented in PyTorch and executed on an NVIDIA GeForce RTX 3080 GPU. The model was trained for 1000 epochs on IP, 500 epochs on PU and UH, and 300 epochs on HC. These epoch settings were determined according to the convergence behavior observed under the fixed sampling protocols, as shown in Fig. 8. All experiments follow the parameter settings listed in Table I, with any exceptions stated separately.

3) Comparison Methods: To demonstrate the effectiveness of the proposed method, SDMCNet is compared with several representative HSI classification methods. For a fair comparison, the hyperparameter settings of the comparison methods follow those reported in their original papers. For SVM [6], the radial basis function (RBF) kernel is adopted. For RF [9], 500 decision trees are used, and $\sqrt{B}$ spectral features are randomly selected as candidate variables at each split node, where B denotes the number of spectral bands. 3D-CNN [40] employs stacked 3-D convolutional blocks with a kernel size of 3 × 3 × 3, while HybridSN [14] combines 3-D and 2-D convolutional layers for spatial–spectral feature extraction. ViT [41] and SpectralFormer [17] use a patch size of 1 × 1 across all datasets. SSTN [42] operates on a 9 × 9 local

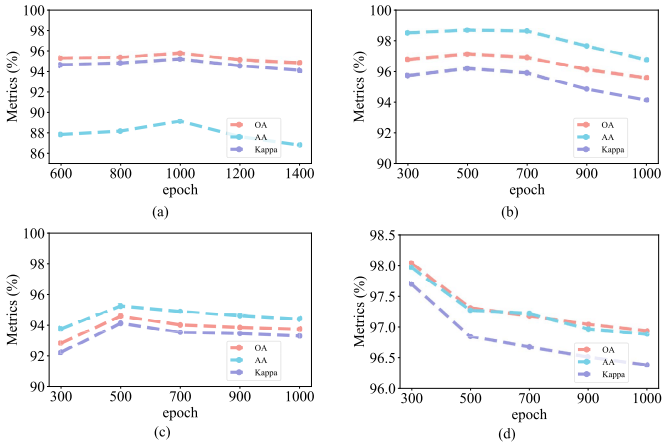


Fig. 8. Epoch sensitivity analysis on four datasets. (a) IPs. (b) PU. (c) Houston 2013. (d) HC.

TABLE I

HYPERPARAMETER CONFIGURATION OF SDMCNET, INCLUDING MODULE-SPECIFIC PARAMETERS, REGULARIZATION WEIGHTS, AND TRAINING SETTINGS

Parameter	Value	Description
w_{loc}	0.7	Fusion coefficient for the multiscale convolution branch in DLRM
w_{mam}	0.3	Fusion coefficient for the Mamba branch in DLRM
η	0.4	Initial value of the residual scaling coefficient in DLRM
z	4	Block size of the block-level row and column serpentine paths in MPSM
s	$\{2, 4\}$	Downsampling scales used in the multiscale correction branch of BCEM
μ	0.8	Initial value of the residual scaling coefficient for the boundary constrained correction in BCEM
ϕ	0.16	High frequency detail compensation gain coefficient in BCEM
θ	0.7	Boundary suppression gating strength coefficient in BCEM
r	$\{1, 2, 4\}$	Dilation rates of the multi-scale dilated convolutions in SPAR
ρ	0.8	Residual scaling coefficient in SPAR
d	3	Anchor sampling stride in SSAG
V	8	Number of candidate sampling points for each anchor in SSAG
R	8	Deformable sampling radius in SSAG
m	5	Number of top-m spectral tokens selected for each anchor
E	4	Number of experts in the MoE module
k	2	Number of activated experts in sparse MoE routing
λ_{ent}	0.003	Weight of the routing entropy regularization term
λ_{moe}	2.5	Weight of the MoE auxiliary loss
λ_{reg}	2	Weight of the SSAG coverage and smoothness regularization term
Optimizer	Adam	Optimizer used for model training
Learning rate	0.0005	Initial learning rate
Batch size	1	Training batch size

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TABLE II

OA, AA, AND KAPPA (%) ACHIEVED BY DIFFERENT METHODS ON THE IP DATASET. THE BEST RESULT IN EACH ROW IS BOLDED

Class	ML-Based		CNN-Based		Transformer-Based				SSM-Based			Ours
	SVM	RF	3D-CNN	HybridSN	ViT	SpectralFormer	SSTN	SSFTT	IGroupSS-Mamba	MambaHSI	MambaLG	SDMCNet
1	8.00	48.00	**100.00**	72.00	96.00	**100.00**	96.00	**100.00**	92.00	96.00	48.00	**100.00**
2	54.51	66.37	90.81	85.19	74.37	77.92	90.66	73.48	85.63	71.41	85.33	**91.41**
3	48.76	52.72	67.07	66.09	66.58	66.33	92.82	78.21	74.75	74.75	93.56	**95.79**
4	12.12	38.38	65.65	37.37	53.53	53.53	69.69	48.48	**71.72**	54.55	49.49	70.71
5	71.16	83.21	87.59	86.13	88.68	86.49	89.78	93.06	**95.26**	90.88	71.53	87.59
6	95.48	95.76	95.48	92.09	96.89	95.76	98.02	90.67	91.81	89.83	**99.15**	98.59
7	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	**100.00**	0.00	0.00	0.00
8	63.60	**100.00**	99.20	96.40	98.40	94.80	**100.00**	**100.00**	**100.00**	91.60	**100.00**	**100.00**
9	20.00	50.00	30.00	80.00	**100.00**	**100.00**	90.00	**100.00**	**100.00**	0.00	0.00	**100.00**
10	38.36	41.94	75.74	82.70	68.19	74.95	88.86	62.02	95.63	61.63	92.05	**97.22**
11	88.73	86.00	84.22	75.59	85.72	87.32	95.68	80.75	82.07	88.26	94.37	**98.87**
12	38.29	47.16	90.42	65.60	75.17	71.63	94.68	69.14	92.20	90.07	93.97	**97.16**
13	92.50	88.75	98.75	95.00	97.50	95.00	87.50	91.25	96.25	97.50	**100.00**	98.75
14	79.26	92.11	97.61	94.31	88.25	90.82	99.81	98.16	**100.00**	95.78	98.90	97.80
15	21.21	55.55	70.70	55.56	79.79	67.67	74.74	30.30	58.59	45.45	76.77	**95.96**
16	31.81	93.18	94.90	86.36	93.18	88.63	77.27	72.72	84.09	65.91	86.36	**95.45**
OA (%)	64.95	73.48	86.15	80.60	81.45	82.38	93.12	79.77	88.18	81.43	90.80	**95.79**
AA (%)	47.74	64.94	78.01	73.15	78.89	78.18	84.09	74.26	88.75	69.60	74.34	**89.08**
Kappa (%)	58.66	69.37	84.25	77.95	78.80	79.88	92.16	77.02	86.61	78.83	89.54	**95.21**

TABLE III

OA, AA, AND KAPPA (%) ACHIEVED BY DIFFERENT METHODS ON THE PU DATASET. THE BEST RESULT IN EACH ROW IS BOLDED

Class	ML-Based		CNN-Based		Transformer-Based				SSM-Based			Ours
	SVM	RF	3D-CNN	HybridSN	ViT	SpectralFormer	SSTN	SSFTT	IGroupSS-Mamba	MambaHSI	MambaLG	SDMCNet
1	74.29	80.56	86.61	94.96	80.71	78.60	96.55	94.79	84.34	**97.41**	94.20	96.89
2	74.17	72.30	88.64	94.61	69.66	72.96	93.88	96.25	85.57	91.39	**96.40**	95.12
3	78.58	72.73	81.69	85.39	66.68	82.99	97.69	91.04	74.19	96.80	**99.50**	98.85
4	95.71	95.10	96.01	98.31	89.27	97.77	98.88	97.33	91.90	95.55	99.09	**99.29**
5	99.27	99.35	99.91	**100.00**	99.60	99.27	99.83	**100.00**	99.44	**100.00**	**100.00**	99.92
6	90.46	78.35	96.22	99.69	92.84	79.44	98.72	97.46	85.86	99.15	99.19	**99.96**
7	94.14	88.94	96.65	**100.00**	92.11	92.43	**100.00**	99.75	83.09	99.43	99.11	99.43
8	84.19	75.15	84.64	92.54	88.30	83.08	85.56	93.85	73.26	94.25	88.67	**98.83**
9	99.88	**100.00**	99.52	96.69	**100.00**	**100.00**	99.88	99.52	99.17	99.41	99.17	**100.00**
OA (%)	80.55	78.03	89.76	95.27	79.11	79.61	95.17	96.07	84.88	94.69	96.30	**97.12**
AA (%)	87.85	84.72	91.88	95.80	86.57	87.40	96.78	96.67	86.31	97.04	97.25	**98.69**
Kappa (%)	75.29	71.99	86.66	93.75	73.58	74.02	93.63	94.78	80.27	93.03	95.10	**96.19**

neighborhood, whereas SSFTT [18] and IGroupSS-Mamba [28] use 13 $\times$ 13 input windows. MambaHSI [31] is configured with $G = 4$ groups and a hidden dimension of $D = 128$. For MambaLG [43], the embedding dimension is set to 64 for most datasets and 84 for the Houston 2013 dataset, following its original configuration.

C. Comparison and Analysis of Experimental Results

Tables II–V report the quantitative results on the IPs, PU, Houston 2013, and HC datasets. Although the relative performance of the representative methods varies across datasets, SDMCNet achieves the highest OA, AA, and Kappa on all four datasets. On the IP and UH datasets, SDMCNet improves OA, AA, and Kappa over the suboptimal SSTN method by 2.67%, 4.99%, and 3.05% on IP and by 2.87%, 2.92%, and 3.12% on UH. SSTN restricts spatial interactions to a fixed 9 $\times$ 9 neighborhood, whereas SDMCNet integrates local spatial relationships with long-range context and propagates information along multiple paths, thereby enabling more effective use of spatial dependencies beyond this fixed neighborhood. The advantage of SDMCNet is reflected not only in aggregate metrics but also in major crop categories with representative parcel structures and relatively high sample proportions. For example, compared with IGroupSS-Mamba, SDMCNet improves the accuracies of Corn-mintill and Soybean-mintill by 21.04% and 16.80%, respectively. On PU, where each class is represented by only 100 labeled training samples, SDMCNet attains an AA of 98.69%, reflecting its ability to learn discriminative class representations under limited supervision. On HC, SDMCNet achieves an OA of 98.04%, an AA of 97.97%, and a Kappa of 97.70% while obtaining the highest accuracy in 12 of the 16 classes. These classes include Red roof, Plastic, and Road, as well as seven vegetation classes. This result indicates that constructing informative spectral groups and coordinating complementary experts allow SDMCNet to distinguish subtle differences among spectrally similar vegetation classes while accounting for spectral variations across different land-cover materials.

From the qualitative results shown in Figs. 9–12, traditional and CNN-based methods commonly produce scattered predictions within homogeneous regions and fragmented contours because their spatial interactions are mainly restricted to local neighborhoods. Transformer and SSM methods improve regional connectivity, but residual confusion remains around small parcels, elongated objects, transitional regions, and densely distributed narrow structures. However, SDMCNet produces cleaner parcel interiors and more complete irregular boundaries for spatially coherent crop categories such as Corn-mintill, Soybean-notill, Soybean-mintill, and Soybean-clean.

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TABLE IV

OA, AA, AND KAPPA (%) ACHIEVED BY DIFFERENT METHODS ON THE UH DATASET. THE BEST RESULT IN EACH ROW IS BOLDED

Class	ML-Based		CNN-Based		Transformer-Based				SSM-Based			Ours
	SVM	RF	3D-CNN	HybridSN	ViT	SpectralFormer	SSTN	SSFTT	IGroupSS-Mamba	MambaHSI	MambaLG	SDMCNet
1	83.38	83.47	83.57	79.67	83.76	**86.70**	81.57	81.19	86.23	80.63	82.34	81.39
2	95.39	98.30	94.64	87.31	97.18	94.64	**100.00**	97.27	95.77	98.59	**100.00**	99.91
3	**100.00**	98.01	98.41	67.92	**100.00**	99.80	99.60	89.30	**100.00**	67.13	99.41	**100.00**
4	98.10	97.63	98.95	75.37	98.67	96.87	96.59	92.99	92.33	96.50	**99.43**	99.24
5	97.15	96.40	**100.00**	**100.00**	96.88	98.29	99.90	**100.00**	98.96	**100.00**	**100.00**	**100.00**
6	95.10	97.20	94.40	90.20	95.10	**99.30**	95.80	95.80	95.10	95.80	95.10	96.50
7	79.75	82.74	90.67	76.86	86.01	89.27	**91.97**	84.23	76.68	82.18	88.99	90.86
8	49.19	40.64	57.07	89.07	54.51	61.82	80.15	55.08	64.20	72.08	85.47	**94.02**
9	72.52	69.59	81.30	73.18	67.99	77.62	85.74	**93.67**	75.83	86.59	86.69	92.45
10	72.87	59.16	69.78	38.41	85.14	81.94	90.92	57.52	75.39	85.33	73.75	**92.28**
11	86.81	76.75	92.78	63.75	64.61	84.91	97.62	96.48	89.85	75.14	95.92	**97.91**
12	53.50	49.56	70.60	88.28	57.73	72.23	**97.89**	88.85	86.55	86.94	87.61	92.03
13	68.77	60.00	89.47	82.10	69.47	72.28	84.21	87.71	73.33	78.60	91.23	**92.63**
14	**100.00**	99.19	99.59	**100.00**	98.79	99.59	**100.00**	**100.00**	99.19	99.60	99.60	99.60
15	98.52	97.46	97.04	**100.00**	98.31	97.67	93.02	98.73	98.10	90.91	**100.00**	**100.00**
OA (%)	80.94	77.68	85.67	78.47	81.24	85.82	91.73	86.02	85.56	85.98	91.11	**94.60**
AA (%)	83.40	80.41	87.88	80.81	83.61	87.53	92.33	87.92	87.17	86.40	92.36	**95.25**
Kappa (%)	79.36	75.85	84.45	76.75	79.68	84.62	91.02	84.86	84.34	84.77	90.35	**94.14**

TABLE V

OA, AA, AND KAPPA (%) ACHIEVED BY DIFFERENT METHODS ON THE HC DATASET. THE BEST RESULT IN EACH ROW IS BOLDED

Class	ML-Based		CNN-Based		Transformer-Based				SSM-Based			Ours
	SVM	RF	3D-CNN	HybridSN	ViT	SpectralFormer	SSTN	SSFTT	IGroupSS-Mamba	MambaHSI	MambaLG	SDMCNet
1	75.20	73.75	61.12	95.92	91.89	76.24	96.12	92.63	70.55	91.50	95.88	**97.32**
2	51.90	40.91	57.65	92.54	66.02	57.33	92.98	91.42	52.22	81.39	95.10	**95.16**
3	75.23	77.95	62.70	95.15	65.72	82.71	98.41	98.60	57.12	97.06	99.54	**99.77**
4	95.71	92.29	94.06	**98.93**	90.45	95.61	95.88	97.65	94.51	96.29	98.31	97.28
5	83.40	83.80	91.60	**100.00**	26.60	78.90	**100.00**	99.40	73.60	99.70	99.90	99.90
6	43.06	34.73	40.89	93.86	16.84	49.43	73.89	95.24	50.31	80.73	96.05	**96.47**
7	87.39	88.58	91.16	97.45	56.96	83.62	97.31	97.54	76.36	95.60	95.67	**98.56**
8	59.90	59.28	39.13	86.75	65.22	56.46	90.34	94.44	62.30	75.57	95.67	**97.71**
9	57.43	47.29	27.12	89.75	59.38	64.98	96.66	95.83	64.92	92.60	99.08	**99.67**
10	93.02	88.39	86.41	99.04	83.93	96.03	99.13	98.41	87.53	97.12	98.57	**99.80**
11	93.94	92.54	86.15	91.29	81.74	85.51	97.15	97.43	79.53	97.87	96.13	**99.10**
12	70.39	69.10	45.67	97.87	23.66	76.45	98.30	99.54	60.22	71.17	99.20	**99.94**
13	53.14	55.94	38.20	72.98	33.89	61.60	82.43	91.13	50.04	81.40	90.02	**91.15**
14	66.06	63.55	67.60	88.52	79.44	67.46	88.22	95.42	69.51	96.29	96.18	**97.14**
15	86.32	80.98	88.78	98.71	49.78	89.74	99.14	**99.78**	84.19	96.58	99.36	99.04
16	89.48	95.56	93.56	96.49	98.59	98.57	**99.61**	98.75	96.30	96.95	98.02	99.52
OA (%)	76.31	75.99	71.03	93.49	79.91	80.06	95.42	95.91	75.90	91.78	96.75	**98.04**
AA (%)	73.85	71.54	66.99	93.45	61.88	76.29	94.10	96.45	70.58	90.49	97.04	**97.97**
Kappa (%)	72.72	72.26	66.66	92.40	76.37	76.84	94.63	95.22	72.11	90.40	96.20	**97.70**

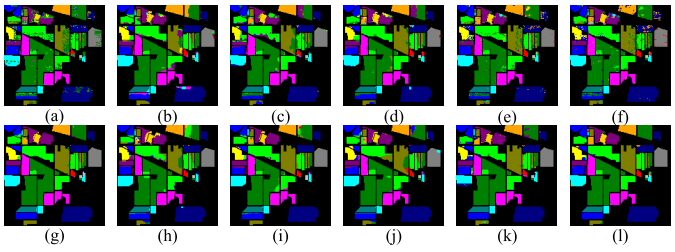


Fig. 9. Qualitative visualization of the classification map for IP dataset. (a) SVM. (b) RF. (c) 3D-CNN. (d) HybridSN. (e) ViT. (f) SpectralFormer. (g) SSTN. (h) SSFTT. (i) IGroupSS-Mamba. (j) MambaHSI. (k) MambaLG. (l) SDMCNet.

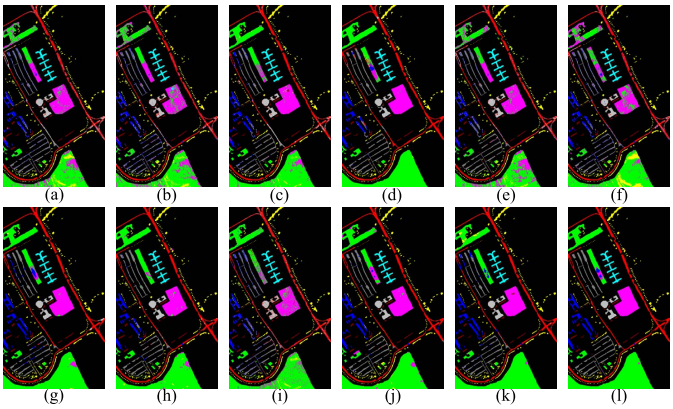


Fig. 10. Qualitative visualization of the classification map for PU dataset. (a) SVM. (b) RF. (c) 3D-CNN. (d) HybridSN. (e) ViT. (f) SpectralFormer. (g) SSTN. (h) SSFTT. (i) IGroupSS-Mamba. (j) MambaHSI. (k) MambaLG. (l) SDMCNet.

Its predictions for Road and Highway retain better continuity, while the elongated cultivated regions in HC contain fewer clustered errors and exhibit more stable field boundaries. In comparison with these methods, the classification maps generated by the proposed method exhibit greater homogeneity within regions, stronger continuity of elongated structures, and more accurate localization of class boundaries, especially in connected parcels and structurally complex regions.

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TABLE VI
ABLATION STUDY OF SDMCNET MODULES ON FOUR DATASETS

The modules of SDMCNet						IP dataset			PU dataset			UH dataset			HC dataset		
SPAR	SSAG	LRDCC	DLRM	MPSM	BCEM	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa	OA	AA	Kappa
✓						83.52	73.42	81.21	86.53	89.11	82.45	83.49	85.46	82.11	78.03	73.37	74.55
✓	✓					84.29	75.04	82.11	87.36	89.23	83.49	85.90	87.63	84.71	79.23	74.78	75.92
✓	✓	✓				84.48	77.65	82.34	87.61	89.35	83.75	86.15	87.76	84.97	79.80	75.09	76.56
✓	✓	✓	✓			91.31	84.86	90.09	94.59	95.77	92.87	89.98	91.53	89.15	94.72	94.40	93.82
✓	✓	✓	✓	✓		93.99	85.62	93.15	96.94	97.97	95.95	93.35	94.30	92.78	97.38	97.59	96.93
✓			✓	✓		93.24	82.70	92.30	95.23	97.20	93.73	92.46	93.35	91.82	95.81	95.74	95.10
✓					✓	90.86	82.94	89.72	93.14	95.01	90.99	89.75	91.24	88.87	89.40	87.61	87.64
✓			✓	✓	✓	94.90	85.39	94.19	96.34	97.93	95.17	93.82	94.46	93.18	96.34	96.46	95.72
✓	✓	✓			✓	92.47	83.86	91.78	94.08	95.56	92.18	91.24	92.36	90.58	91.86	90.74	90.58
✓	✓	✓	✓	✓	✓	**95.79**	**89.08**	**95.21**	**97.12**	**98.69**	**96.19**	**94.60**	**95.25**	**94.14**	**98.04**	**97.97**	**97.70**

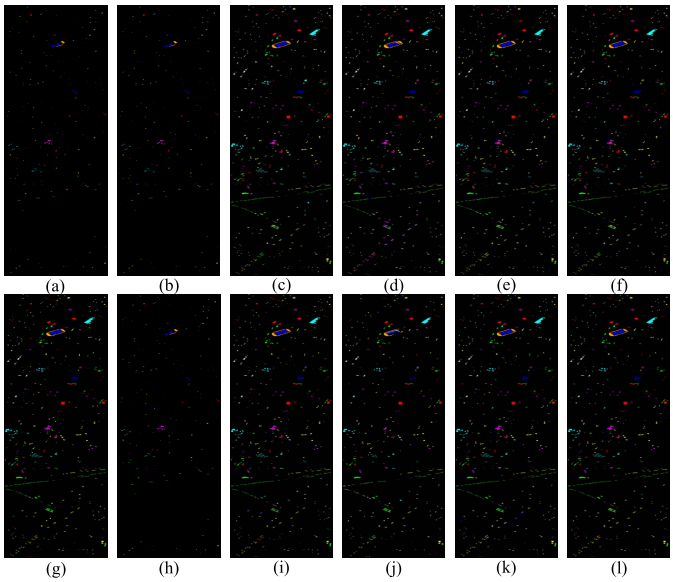


Fig. 11. Qualitative visualization of the classification map for Houston 2013 dataset. (a) SVM. (b) RF. (c) 3D-CNN. (d) HybridSN. (e) ViT. (f) SpectralFormer. (g) SSTN. (h) SSFTT. (i) IGroupSS-Mamba. (j) MambaHSI. (k) MambaLG. (l) SDMCNet.

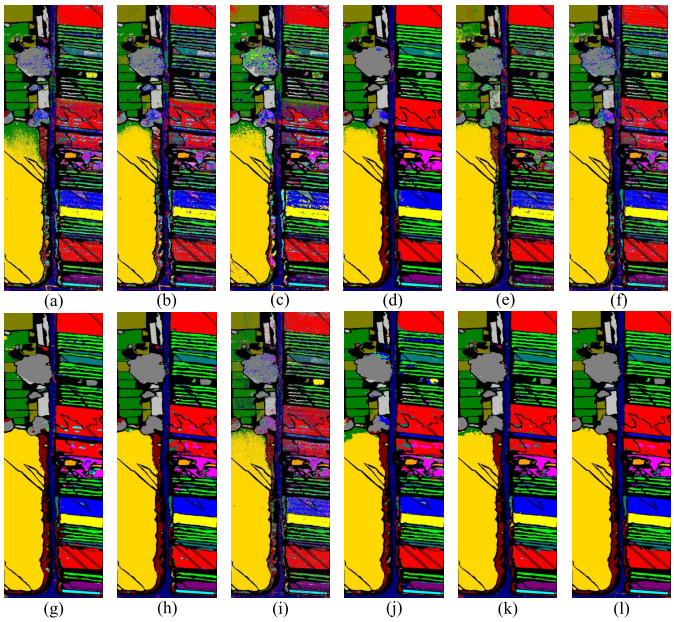


Fig. 12. Qualitative visualization of the classification map for HC dataset. (a) SVM. (b) RF. (c) 3D-CNN. (d) HybridSN. (e) ViT. (f) SpectralFormer. (g) SSTN. (h) SSFTT. (i) IGroupSS-Mamba. (j) MambaHSI. (k) MambaLG. (l) SDMCNet.

At the class level, several performance differences warrant further examination. Grass-pasture is often adjacent to Grass-trees or located in transitional vegetation regions. In these areas, neighboring pixels provide less reliable contextual information, and the coarse 20-m spatial resolution further increases mixed pixel interference. Parking Lots 1 and 2 contain 1041 and 285 test samples, respectively. The larger test set for Parking Lot 1 includes more spatial instances with varying pavement conditions, markings, shadows, vehicle presence, and fragmented layouts. These variations also increase its overlap with Highway, Commercial, and Parking Lot 2 in both spectral responses and spatial structures. By emphasizing localized cues more directly, the fixed 13 $\times$ 13 neighborhood used by IGroupSS-Mamba and the 9 $\times$ 9 neighborhood used by SSTN may contribute to their higher accuracies on Grass-pasture and Parking Lot 1, respectively. However, SDMCNet ranks second on Parking Lot 1 and achieves the highest accuracy on Parking Lot 2. For Healthy grass, discrimination from related vegetation categories mainly depends on subtle spectral variations associated with vegetation condition. SpectralFormer achieves 86.70% on this category, compared with 81.39% for SDMCNet. This difference may be associated with the 1 $\times$ 1 patch used by SpectralFormer, which limits the influence of spatial information from neighboring pixels. Taken together, these observations suggest that the remaining class-level performance differences reflect how each method models class-specific spatial and spectral properties, while SDMCNet maintains strong overall performance across the evaluated datasets.

D. Ablation Study

To validate the effectiveness of the individual modules in SDMCNet, we conduct ablation studies using SPAR as the baseline. Table VI reports OA, AA, and Kappa on the IP, PU, UH, and HC datasets. The consistent performance gains show that the modules contribute to classification from complementary perspectives, although their effects differ. SSAG and LRDCC mainly strengthen class-balanced spectral discrimination by suppressing redundant band responses and

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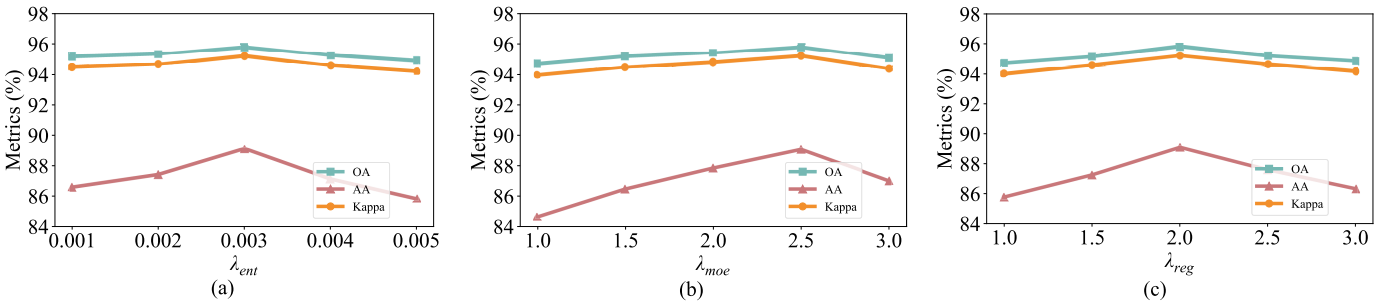


Fig. 13. Sensitivity analysis of regularization weights on the IP dataset. (a) λ_{ent}. (b) λ_{moe}. (c) λ_{reg}. The remaining weights are fixed during each analysis.

TABLE VII
ABLATION STUDY OF DIFFERENT LOSS REGULARIZATION TERMS ON THE IP DATASET

Loss terms				IP dataset		
CE	L_{ent}	L_{moe}	L_{reg}	OA	AA	Kappa
✓				95.11	86.60	94.44
✓	✓			95.28	86.85	94.63
✓		✓		95.45	87.61	94.83
✓			✓	95.41	87.36	94.77
✓		✓	✓	95.56	88.63	94.94
✓	✓	✓	✓	**95.79**	**89.08**	**95.21**

adapting the representation to heterogeneous spectral patterns. This contribution is particularly evident in the pronounced AA improvement on IP. DLRM and MPSM account for most of the OA gain through enhanced local structural continuity and long-range spatial interaction. Their advantage is apparent for the densely distributed small targets in UH and for the elongated field structures and complex transition regions in HC. BCEM further improves boundary-sensitive classification through multiscale correction and constrained feature updates near class contours. Its effectiveness with SPAR alone and within the complete spatial branch confirms that boundary correction provides an independent contribution while remaining complementary to contextual spatial modeling. The combination results also show that the spatial modules primarily establish region-level consistency, whereas SSAG and LRDCC further resolve residual confusion caused by subtle spectral differences and class imbalance. The complete model achieves the best performance across all four datasets, indicating that its advantage does not arise from simple module accumulation but from the complementary effects of spectral discrimination, spatial dependency modeling, and boundary correction.

IV. DISCUSSION

In this section, we first analyze the sensitivity of key hyperparameters to examine the robustness of the proposed modules under different parameter settings. Then, we further evaluate the effectiveness of SDMCNet in class boundary identification and spectral feature calibration for HSI classification.

A. Effect of Loss Regularization Terms

Table VII reports the loss ablation results on the IP dataset. The CE-only baseline provides direct pixel-level supervision, but uncertainty in expert routing and instability in spectral anchor grouping are still not effectively constrained, potentially causing imbalanced expert utilization and redundant spectral sampling, particularly for minority classes and spectrally mixed regions. All configurations incorporating L_{ent}, L_{moe}, or L_{reg} improve upon this baseline. Combining L_{moe} and L_{reg} further improves AA, demonstrating their complementary benefits for class-balanced discrimination, while adding L_{ent} yields the best OA, AA, and Kappa by reducing routing uncertainty. Because the contribution of each term depends on its weight, Fig. 13 examines λ_{ent}, λ_{moe}, and λ_{reg}. Insufficient weights weaken regularization, whereas excessive weights may overemphasize the auxiliary objectives and interfere with classification optimization. The setting $\lambda_{\text{ent}} = 0.003$, $\lambda_{\text{moe}} = 2.5$, and $\lambda_{\text{reg}} = 2.0$ therefore balances routing confidence, expert collaboration, and spectral grouping flexibility.

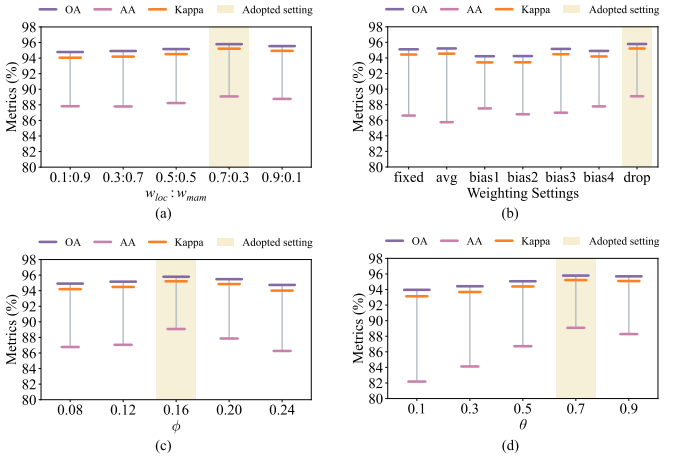


Fig. 14. Parameter configuration analysis of DLRM, MPSM, and BCEM on the IP dataset.

B. Parameter Configuration Analysis of Spatial Modules

The key parameter configurations of DLRM, MPSM, and BCEM are examined on the IP dataset, with the results summarized in Fig. 14.

As shown in Fig. 14(a), DLRM achieves the best OA, AA, and Kappa at $w_{\text{loc}} : w_{\text{mam}} = 0.7 : 0.3$. This configuration emphasizes local multiscale refinement for preserving fine spatial structures while retaining complementary context from the Mamba branch. Further increasing the local-branch weight to 0.9 overemphasizes local responses and limits the

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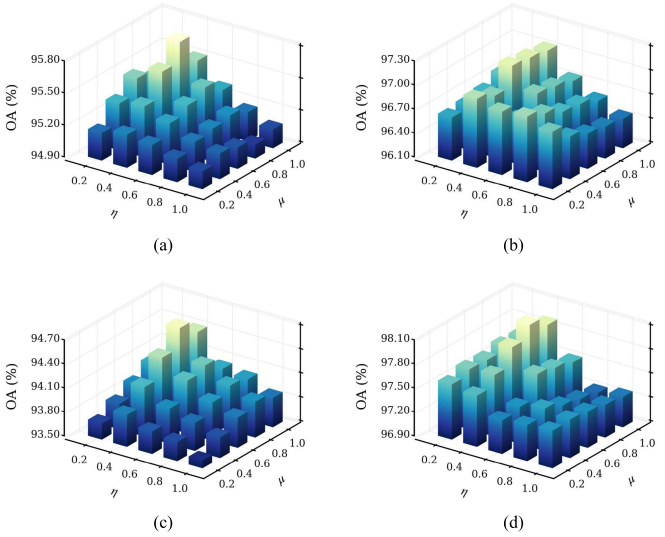


Fig. 15. Analysis of the initialization values of the residual scaling coefficients η and μ on four hyperspectral datasets. (a) IPs. (b) PU. (c) Houston 2013. (d) HC.

complementary role of the Mamba branch, thereby reducing all three metrics.

To investigate the effect of multipath fusion on directional bias, seven methods are compared in Fig. 14(b). Randomly removing one path achieves the best OA, AA, and Kappa. By varying the available path combinations during training, this strategy reduces dependence on fixed directional preferences and promotes the complementary use of spatial information from the remaining directions.

The sensitivity of BCEM to the detail residual gain ϕ and boundary suppression coefficient θ is presented in Fig. 14(c) and (d), respectively. The best performance is obtained at $\phi = 0.16$ and $\theta = 0.70$. Under this configuration, weak boundary structures are reinforced, while the correction is concentrated around relevant boundary regions, reducing interference from adjacent classes and improving boundary discrimination.

C. Analysis of the Initialization Values of Residual Scaling Coefficients

Fig. 15 examines the initialization values of the learnable residual scaling coefficients η in DLRM and μ in BCEM. Although both coefficients are updated during training, their initial values determine the relative strength of spatial refinement and boundary constrained correction at the beginning of optimization. An excessively small η may limit the fused refinement provided by DLRM, whereas an overly large value may cause the refined response to dominate the original features and interfere with subsequent spatial modeling. Likewise, a small μ weakens the contribution of the boundary constrained correction, while an excessively large value may overemphasize the gated correction relative to Y_{MPSM}, leading to overcorrection and reduced spatial stability. Across the four datasets, $\eta = 0.4$ and $\mu = 0.8$ provide consistently favorable performance. These initialization values are therefore used in all subsequent experiments.

TABLE VIII

COMPARISON OF EDGE SOURCE AGGREGATION STRATEGIES FOR BOUNDARY CORRECTION ON THE HC DATASET

Edge aggregation	OA	AA	Kappa	Boundary OA
Channel averaging	**98.04**	**97.97**	**97.70**	**92.72**
Learnable channel weighting	97.81	97.75	97.43	92.09
Max response aggregation	97.45	97.35	97.01	91.60
Energy based aggregation	97.16	97.03	96.68	90.84

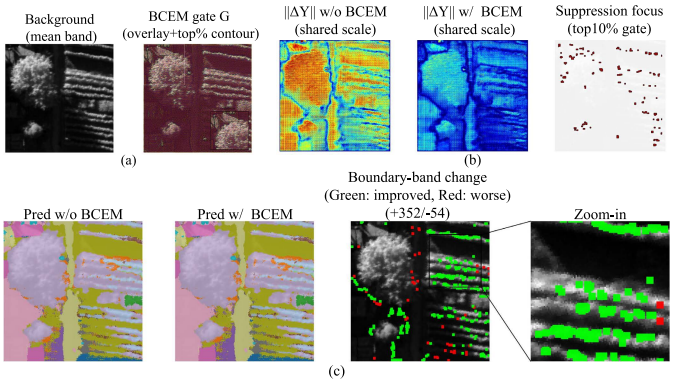


Fig. 16. Boundary sensitivity analysis of BCEM on the HC dataset.

D. Effect of Edge Source Aggregation on Boundary Correction

Different channel aggregation rules produce different spatial source maps for subsequent boundary extraction. In BCEM, channel averaging reduces the multichannel feature representation to a single-channel spatial source map, which is subsequently processed by the fixed Laplacian operator and the boundary gate to regulate multiscale feature correction near class boundaries. We therefore compare channel averaging with learnable feature channel weighting, max response aggregation, and energy-based aggregation while keeping the remaining network settings unchanged. As shown in Table VIII, channel averaging achieves the highest boundary OA of 92.72%, exceeding the other three aggregation strategies by 0.63%, 1.12%, and 1.88%. This comparison suggests that the spatial source map generated by channel averaging is more suitable for subsequent boundary correction under the current BCEM design.

E. Class Boundary Identification Analysis

As shown in Fig. 16, BCEM improves boundary representation and prediction consistency by coupling boundary-aware gating with constrained feature updates. The gate map in Fig. 16(a) produces strong structured responses around class boundaries and texture discontinuities while remaining weak in homogeneous regions, indicating that BCEM concentrates feature enhancement and information integration on boundary regions most prone to class confusion. Guided by these responses, BCEM suppresses unstable feature variations and limits their propagation to surrounding regions, as illustrated in Fig. 16(b). This targeted boundary refinement leads to the quantitative improvements in Fig. 16(c), where region of interest (ROI) OA increases from 0.8037 to 0.8147 and

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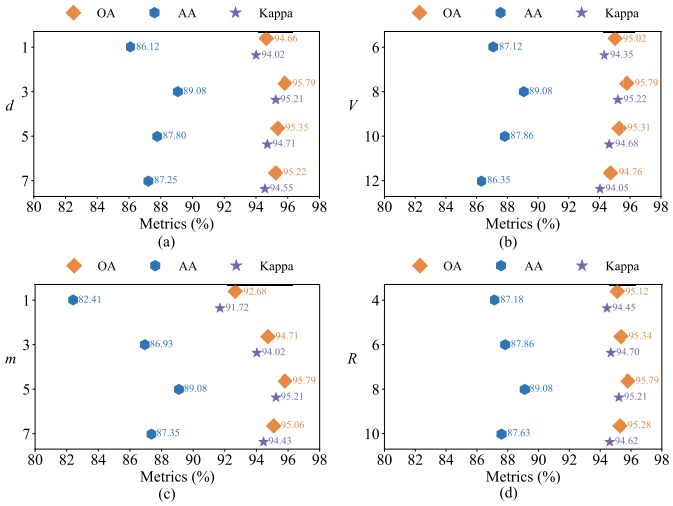


Fig. 17. Parameter analysis of anchor routed spectral grouping on the IP dataset. (a) Anchor stride d. (b) Candidate number V. (c) Selected token number m. (d) Sampling radius R.

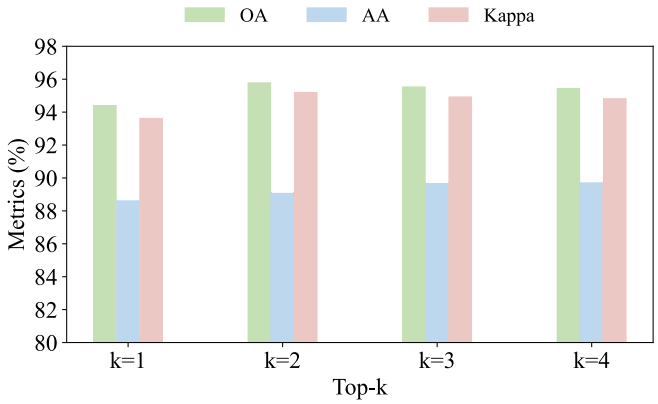


Fig. 18. Impact of the Top-k number of selected experts on the IP dataset.

boundary OA rises from 0.6952 to 0.7095. The zoomed regions further show that these gains are concentrated around elongated boundaries and fragmented structures, demonstrating that BCEM improves boundary accuracy and structural continuity.

F. Parameter Analysis of Anchor Routed Spectral Grouping

Fig. 17 analyzes the sensitivity of the anchor routed spectral grouping parameters on the IP dataset. The best performance is obtained at $d = 3$, $V = 8$, $m = 5$, and $R = 8$, indicating that effective grouping requires a balance among spectral coverage, sampling flexibility, and group compactness. Insufficient anchor coverage or restricted sampling may miss shifted discriminative responses, whereas excessive overlap, sampling range, or token retention may introduce redundant bands. The relatively small performance variations under neighboring settings further indicate that the grouping strategy remains stable within a reasonable parameter range.

G. Analysis of Expert Configuration and Sparse Routing

The subtle spectral differences among crop classes in the IP dataset make it suitable for evaluating the expert combinations and routing behavior of LRDCC. In Table IX, all included experts are activated in settings with one or two experts, whereas Top-2 routing is employed in settings with three or four experts. Graph performs best when used alone, while the performance gains obtained by pairing DCT with either Graph or Mamba demonstrate the complementary role of DCT. Although the combination of Graph and DCT achieves competitive OA, its AA is 1.30% lower than that of the combination of Mamba, Graph, and DCT and 2.18% lower than that of the four-expert combination. The four-expert combination achieves the best OA, AA, and Kappa, further confirming the distinct yet complementary roles of the four experts. Fig. 18 shows that Top-2 routing provides the best overall performance, improving OA and Kappa over $k = 1$ by 1.38% and 1.57%, respectively. Increasing k beyond 2 slightly improves AA but reduces OA and Kappa. With $k = 2$ fixed, Table X shows that expanding the candidate expert pool provides limited additional gains. Although $E = 16$ achieves the highest accuracy, it improves OA, AA, and Kappa over $E = 4$ by only 0.17%, 0.34%, and 0.20%, respectively, while increasing the parameter count from 0.744 to 1.092 M. Therefore, a four-expert candidate pool with Top-2 routing provides a favorable balance among expert complementarity, classification performance, and model complexity.

TABLE IX

PERFORMANCE COMPARISON OF DIFFERENT EXPERT CONFIGURATIONS IN THE MoE BLOCK ON THE IP DATASET. MAMBA, GRAPH, DCT, AND LOWRANK DENOTE THE FOUR EXPERT TYPES USED IN MoE

Mamba	Graph	DCT	LowRank	OA	AA	Kappa
✓				93.92	84.62	93.09
	✓			95.30	86.50	94.65
		✓		91.70	83.40	90.55
			✓	93.60	83.20	92.70
✓	✓			94.85	86.80	94.10
✓		✓		94.25	85.35	93.45
✓			✓	94.90	86.40	94.20
	✓	✓		95.35	86.90	94.70
	✓		✓	94.55	85.30	93.80
		✓	✓	92.70	83.30	91.70
✓	✓	✓		95.05	88.20	94.35
✓	✓		✓	94.95	86.40	94.20
✓		✓	✓	94.05	85.20	93.20
	✓	✓	✓	94.95	86.30	94.15
✓	✓	✓	✓	**95.79**	**89.08**	**95.21**

TABLE X

SENSITIVITY ANALYSIS OF THE NUMBER OF CANDIDATE EXPERTS E IN LRDCC ON THE IP DATASET

E	OA(%)	AA(%)	Kappa(%)	Params(M)
4	95.79	89.08	95.21	**0.744**
8	95.43	88.47	94.81	0.860
12	95.18	88.12	94.56	0.976
16	**95.96**	**89.42**	**95.41**	1.092

H. Effect of Spectral Expert Modeling

To analyze the respective modeling roles of the four experts under identical input conditions, some SSAG spectral groups are extracted from the IP test samples and fed into the MoE block, from which the representations specific to each expert and the corresponding routing weights are obtained.

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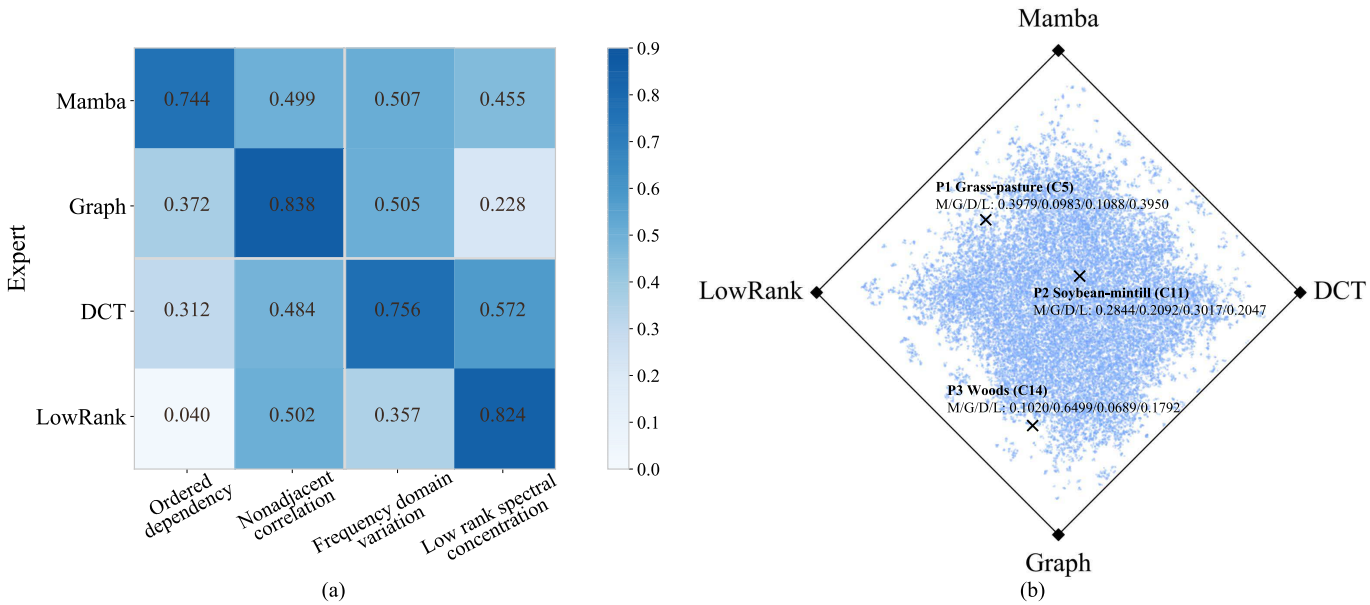


Fig. 19. Spectral expert behavior on the IP dataset. (a) Relative associations between expert representations and spectral characteristics. (b) Coordination of the four experts through adaptive routing.

TABLE XI

TRAINING TIME, TEST TIME, FLOPS, AND NUMBER OF PARAMETERS REQUIRED BY DIFFERENT METHODS WHEN PROCESSING THE IP, PU, UH, AND HC DATASETS

Dataset	Metrics	ML-Based		CNN-Based		Transformer-Based				SSM-Based			Ours
		SVM	RF	3D-CNN	HybridSN	ViT	SpectralFormer	SSTN	SSFTT	IGroupSS-Mamba	MambaHSI	MambaLG	SDMCNet
Indian Pines	Train(s)	3.642	1.380	3.381	15.699	1.523	1.665	0.942	0.585	12.947	0.207	0.086	0.342
	Test(s)	6.604	0.074	3.358	3.889	1.749	1.902	0.947	0.243	2.550	0.029	0.053	0.148
	Flops(G)	–	–	4,188.864	2,335.885	378.948	705.668	73.411	479.476	313.850	4.670	5.616	4.898
	Params(M)	–	–	0.047	5.123	0.346	0.346	0.020	0.153	0.142	0.426	0.193	0.744
Pavia University	Train(s)	0.034	0.432	0.412	2.944	0.122	0.168	0.136	0.138	2.428	1.358	0.491	1.849
	Test(s)	1.737	0.397	18.596	34.387	7.479	7.988	5.004	3.374	10.766	0.468	0.168	0.601
	Flops(G)	–	–	21,538.895	20,763.568	1,934.176	2,798.314	585.325	4,729.769	3,095.907	41.040	51.264	46.971
	Params(M)	–	–	0.047	5.122	0.162	0.162	0.016	0.153	0.142	0.412	0.187	0.736
Houston 2013	Train(s)	0.243	0.781	1.326	7.587	0.537	0.612	0.685	0.157	7.385	4.609	1.964	7.387
	Test(s)	2.056	0.159	78.014	10.040	38.638	42.238	19.335	9.688	8.005	1.190	0.636	1.607
	Flops(G)	–	–	95,499.293	6,047.713	8,644.602	14,024.549	2,025.817	15,161.827	9,924.421	70.980	264.717	190.467
	Params(M)	–	–	0.047	5.123	0.226	0.226	0.018	0.153	0.142	0.418	0.297	1.023
HanChuan	Train(s)	1.292	1.133	2.694	9.052	1.276	1.395	2.861	0.159	6.628	2.269	1.002	3.056
	Test(s)	188.627	3.019	81.011	210.032	48.015	51.614	11.005	5.380	79.183	0.741	0.464	1.104
	Flops(G)	–	–	100,464.517	126,106.068	9,092.963	19,814.425	1,504.144	8,409.387	5,504.513	44.860	99.713	87.682
	Params(M)	–	–	0.047	5.123	0.563	0.563	0.024	0.153	0.142	0.435	0.198	0.748

The same prediction procedure is applied to each expert representation to obtain its association scores with the four spectral characteristics. Fig. 19(a) shows that each expert has its strongest association with the spectral characteristic corresponding to its architectural design. Specifically, Mamba models ordered spectral dependencies through SSM-based sequential modeling. Graph uses dynamically learned connections to aggregate correlated responses across nonadjacent bands. DCT maps spectral responses into the frequency domain to characterize variations at different frequency scales. LowRank reduces redundancy in spectral responses while preserving the dominant global dependencies across bands. These results confirm that the four experts capture the corresponding spectral characteristics through their respective modeling mechanisms, thereby validating the effectiveness of the expert design.

Based on these differentiated modeling roles, Fig. 19(b) maps the routing weight vector (q_M, q_G, q_D, q_L) of each SSAG spectral group to $(x, y) = (q_D - q_L, q_M - q_G)$. The projected points extend toward all four-expert directions, revealing diverse routing preferences across spectral groups. At P1, Mamba captures the ordered spectral progression from chlorophyll absorption in the red region through the red edge to high reflectance in the near-infrared region, while LowRank reduces redundancy among correlated band responses associated with variations in canopy density. P2 is characterized by smooth and localized variations arising from the joint contributions of the soybean canopy and soil background. DCT separates these variations through frequency domain decomposition, while Mamba models their sequential dependencies along the wavelength dimension. In P3, Graph links responses across widely separated spectral regions, while LowRank reduces redundancy within each region. Taken together, these cases demonstrate that the routing mechanism is capable of coordinating complementary experts according to the spectral structure of each group.

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I. Complexity Analysis

Table XI reports the training time, test time, FLOPs, and number of parameters of different methods under the same experimental settings. From the perspective of FLOPs, compared with the deep learning methods across the four datasets, SDMCNet is second only to MambaHSI but better than the remaining compared models. In contrast, the parameter number and runtime results show a different trend. Compared with several baselines, SDMCNet has a larger parameter number and requires longer training and testing times. This is mainly because the proposed method adopts adaptive spatial–spectral modeling to capture more detailed feature relationships rather than using only fixed feature transformations. In particular, the model needs to reorganize spatial–spectral features, select informative spectral responses, and aggregate complementary representations according to the input content. These processes provide more flexible representation capability for mixed pixels, subtle spectral differences, complex spatial boundaries, and imbalanced classes while also introducing additional routing, selection, and aggregation overhead in practical implementation. Overall, SDMCNet maintains relatively low FLOPs and obtains stronger fine-grained spatial–spectral discrimination capability by sacrificing a certain amount of inference time and parameter overhead. In future work, we will further explore lightweight designs to improve the efficiency and deployment feasibility of SDMCNet.

V. Conclusion

In this article, we propose SDMCNet, a spatial–spectral decoupled Mamba–MoE collaborative network for HSI classification with whole-image input. In the spatial branch, PSPFE enhances spatial context modeling, effectively mitigating the directional bias introduced by scanning based on fixed windows while preserving spatial boundary details. In the spectral branch, ARESA employs discriminative spectral grouping and coordinates complementary experts according to spectral characteristics, thereby enabling the model to obtain fine-grained spectral representations under conditions of complex spectral distributions and spectral redundancy. Spectral channel weighting then uses these representations to recalibrate spatial features, thus achieving synergistic spatial–spectral modeling. Extensive experiments on four hyperspectral datasets demonstrate that SDMCNet outperforms other representative methods. Future work will focus on lightweight spatial–spectral collaborative designs to improve computational efficiency and deployment feasibility.

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